Eksamen	**Exam**
Kopiereg voorbehou	Copyright reserved
Elektrisiteit en Elektronika EBN 122	**Electricity and Electronics EBN 122**
13 November 2018	13 November 2018

Toetsinligting - Test Information			
Maksimum punte: Maximum marks:	90	Volpunte: Full marks:	93
Duur van vraestel: Duration of paper:	180 min + (20 for OCR)	Oop-/Toeboek: Open/Closed book:	Toe/Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (this page included)		31	

Important

1. The examination regulations of the University of Pretoria apply.
2. All questions must be answered on the Optical Character Recognition (OCR) form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Belangrik

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
4. Indien van toepassing, moet antwoorde eenhede insluit.
5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
6. Gebruik ten minste 3 beduidende syfers om irrasionele getalle voor te stel.

Lecturers/Dosente:	D. le Roux, J.P. de Villiers, J.P Jacobs

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 180 minutes of the exam. (0-180 min)

Do step 2 within the following 10 minutes. (180-200 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

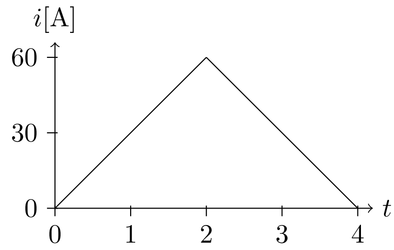
STEP 1 (0 min to 180 min): Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.

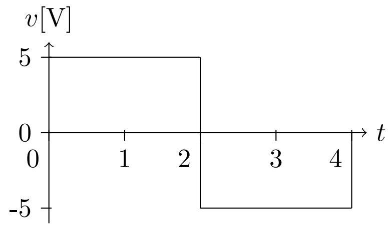
STEP 2 (180 min to 200 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN122E1**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

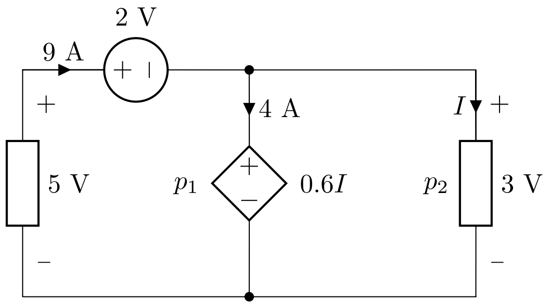
Section 1 / Afdeling 1

Questions 1 to 2 [3]		**Vrae 1 tot 2 [3]**	
The figure below shows the current through and the voltage across a circuit element.		Die figuur hieronder toon die stroom deur en die spanning oor 'n kringbaanelement.	
1	Determine the power delivered to the element at time $t = 3$s. [1]	**1**	Bepaal die drywing gelewer aan die element by tyd $t = 3$s. [1]
2	Find the total energy absorbed by the element for the period of $0 < t < 4$ s. [2]	**2**	Bepaal die totale energie geabsorbeer deur die element vir die tydsduur $0 < t < 4$ s [2]

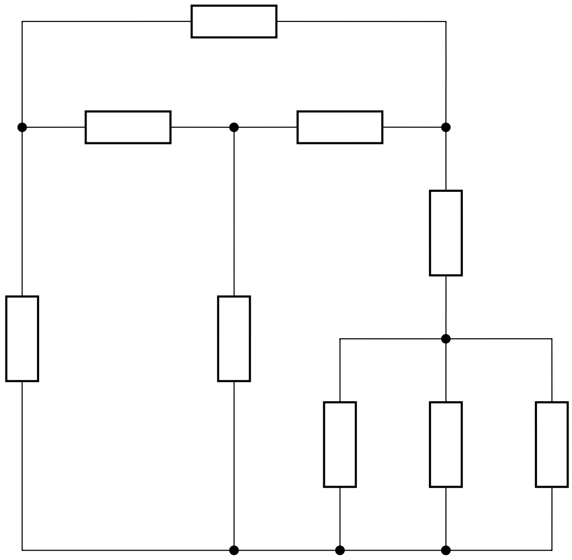




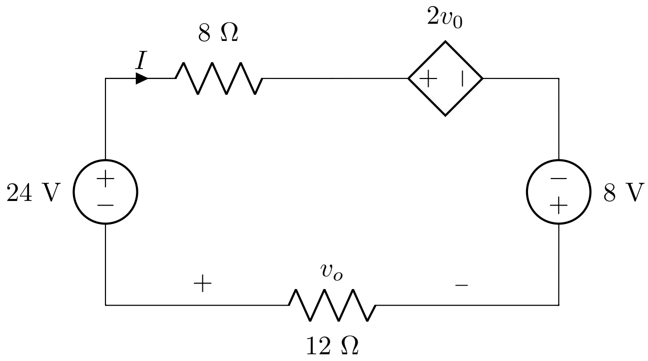
Questions 3 to 4 [2]		Vrae 3 tot 4 [2]	
Assuming $I = 5$ A, calculate the power relating to the following elements in the circuit below:		Indien aanvaar word dat $I = 5$ A, bereken die drywing geassosieer met die volgende elemente in die onderstaande stroombaan:	
3	p_1 [1]	**3**	p_1 [1]
4	p_2 [1]	**4**	p_2 [1]



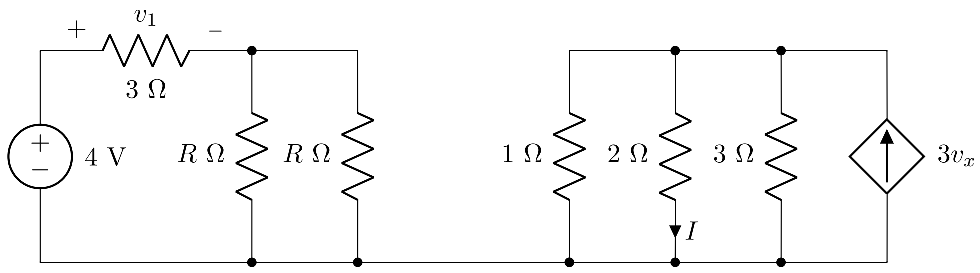
Questions 5 to 7 [3]		**Vrae 5 tot 7 [3]**	
Determine the number of nodes, branches, and independent loops in the circuit diagram below.		Bepaal die hoeveelheid nodes, takke, en onafhanklike lusse in die kringbaan hieronder.	
5	Nodes n [1]	**5**	Nodusse n [1]
6	Branches b [1]	**6**	Takke b [1]
7	Independent loops l [1]	**7**	Onafhanklike lusse l [1]



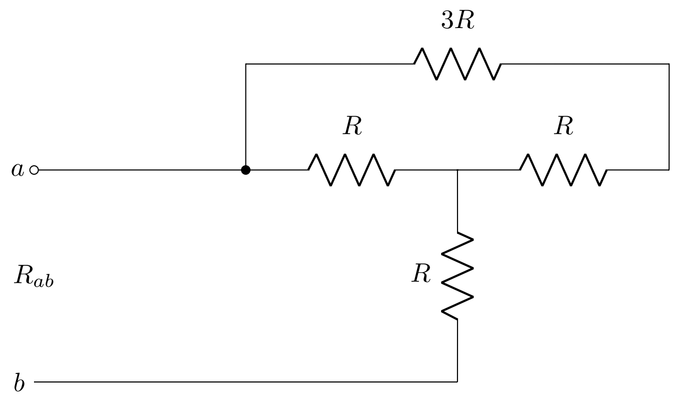
Questions 8 to 9 [2]		**Vrae 8 tot 9 [2]**	
Consider the circuit below. Determine:		Beskou die kringbaan hieronder. Bepaal	
8	I [1]	**8**	I [1]
9	v_0 [1]	**9**	v_0 [1]



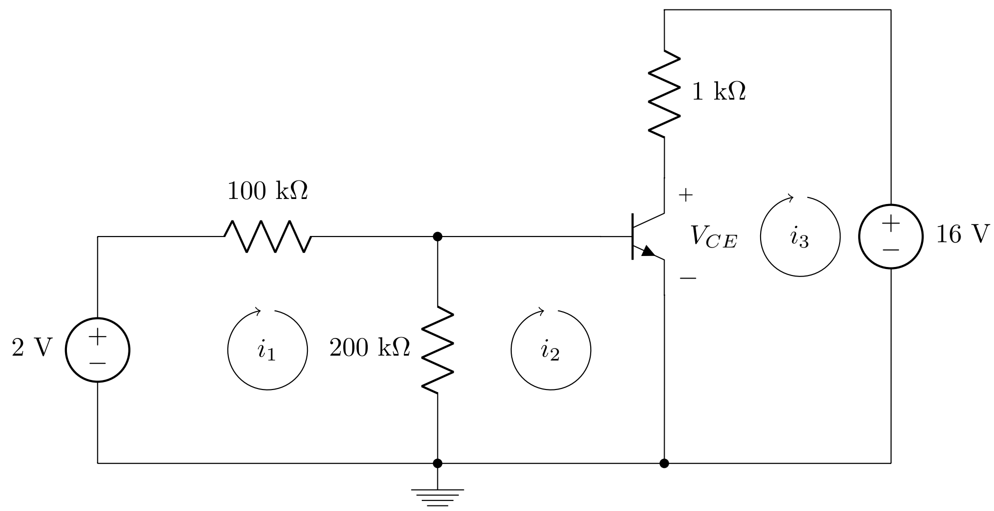
Questions 10 to 11 [3]		**Vrae 10 tot 11 [3]**	
Consider the circuit below.		Beskou die kringbaan hieronder.	
10	If $v_1 = 3$ V, determine R. [2]	**10**	Indien $v_1 = 3$ V, bepaal R. [2]
11	If $v_x = 4$ V, determine I. [1]	**11**	Indien $v_x = 4$ V, bepaal I. [1]



Question 12 [2]		**Vraag 12 [2]**	
12	Determine R if $R_{ab} = 18\ \Omega$.	**12**	Bepaal R indien $R_{ab} = 18\ \Omega$.

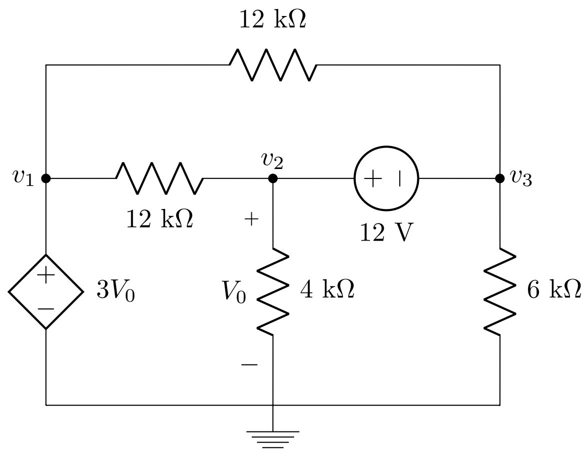


Questions 13 to 17 [5]				**Vrae 13 tot 17 [5]**			
Consider the circuit below containing a transistor with $\beta = 150$ and $V_{BE} = 0.7$ V. Use mesh analysis to set up the following two equations of the form: $$ai_1 + bi_2 = 2 \times 10^{-5}$$ $$ci_1 + di_2 = -0.7 \times 10^{-5}$$ Calculate the unknown coefficients.				Beskou die kringbaan hieronder wat 'n transistor bevat met $\beta = 150$ en $V_{BE} = 0.7$ V. Gebruike lusanalise om die volgende twee vergelykings op te stel met die vorm: $$ai_1 + bi_2 = 2 \times 10^{-5}$$ $$ci_1 + di_2 = -0.7 \times 10^{-5}$$ Bepaal die onbekende koëffisiënte.			
13	a [1]	**14**	b [1]	**15**	c [1]	**16**	d [1]
17	Determine the value of V_{CE} if $i_2 = 9.5$ μA. [1]			**17**	Bepaal die waarde van V_{CE} indien $i_2 =$ 9.5 μA. [1]		

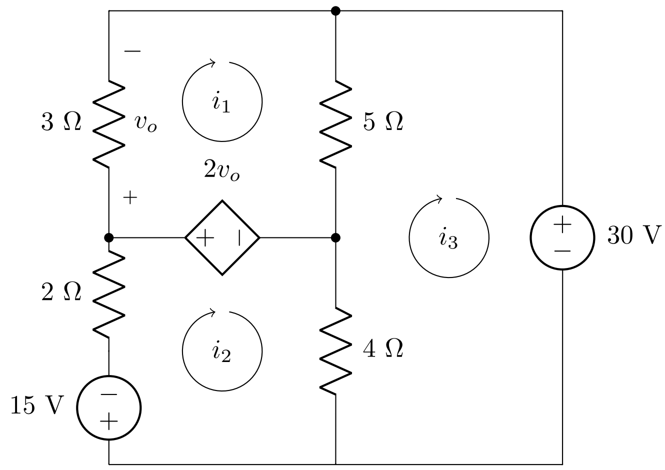


Questions 18 to 21 [4]	**Vrae 18 tot 21 [4]**
Use nodal analysis to set up the following two equations of the form: $$\boldsymbol{e}v_2 + 3v_3 = \boldsymbol{f}$$ $$\boldsymbol{g}v_2 + \boldsymbol{h}v_3 = 12$$ Calculate the unknown coefficients.	Gebruike nodale analise om die volgende twee vergelykings op te stel met die vorm: $$\boldsymbol{e}v_2 + 3v_3 = \boldsymbol{f}$$ $$\boldsymbol{g}v_2 + \boldsymbol{h}v_3 = 12$$ Bepaal die onbekende koëffisiënte.

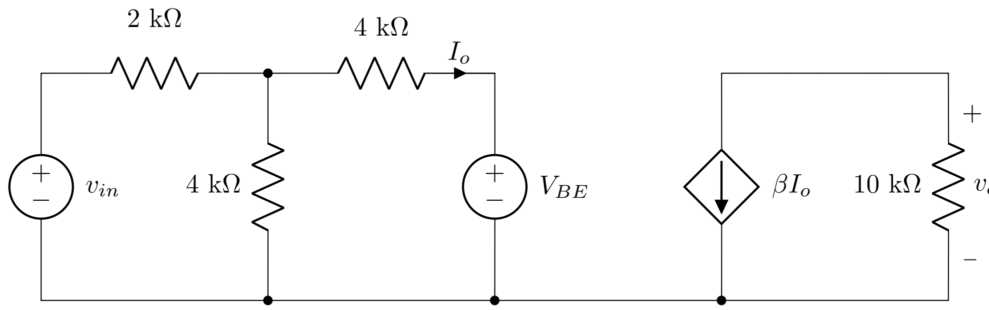
18	e [1]	**19**	f [1]	**20**	g [1]	**21**	h [1]



Questions 22 to 25 [4]				**Vrae 22 tot 25 [4]**			
Use inspection to compile a matrix equation of the form below $$\begin{bmatrix} 2 & k & -5 \\ l & 6 & -4 \\ -5 & p & q \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -15 \\ -30 \end{bmatrix}$$ Then give the values of the following matrix entries.				Gebruike inspeksie om die volgende twee matriksvergelykings op te stel met die vorm: $$\begin{bmatrix} 2 & k & -5 \\ l & 6 & -4 \\ -5 & p & q \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -15 \\ -30 \end{bmatrix}$$ Verskaf dan die waardes van die volgende matrikselemente.			
22	k [1]	**23**	l [1]	**24**	p [1]	**25**	q [1]



Questions 26 to 27 [3]		**Vrae 26 tot 27 [3]**	
Consider a transistor amplifier equivalent circuit below. $V_{BE} = 0.7$ V is the base to emitter voltage, v_{in} is the input voltage and v_{out} is the output voltage. Assume active region operation.		Beskou 'n transistorversterker ekwivalente stroombaan hieronder. $V_{BE} = 0.7$ V is die basis na emitter spanning, v_{in} is die insetspanning en v_{out} is die uitsetspanning. Aanvaar aktiewe gebied werking.	
26	Assume that $\beta = 50$. Determine I_o if $v_{in} =$ 5 V. [1]	**26**	Aanvaar dat $\beta = 50$. Bepaal I_o if $v_{in} = 5$ V? [1]
27	Now let $\beta = 100$. What is the output voltage v_{out}, if $v_{in} = 5$ V? [2]	**27**	Nou laat $\beta = 100$. Wat is die uitsetspanning v_{out}, indien $v_{in} = 5$ V? [2]



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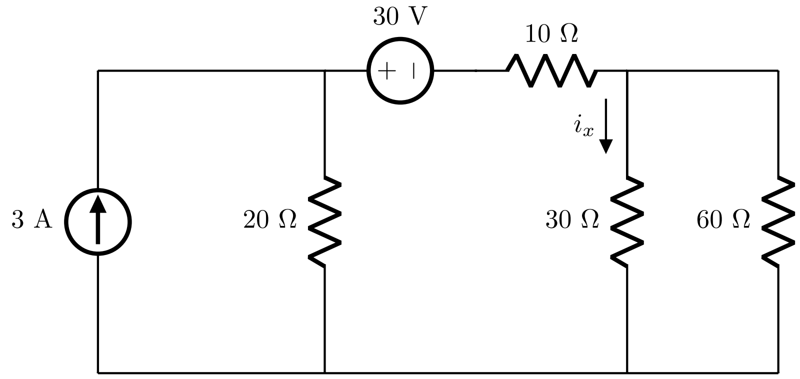
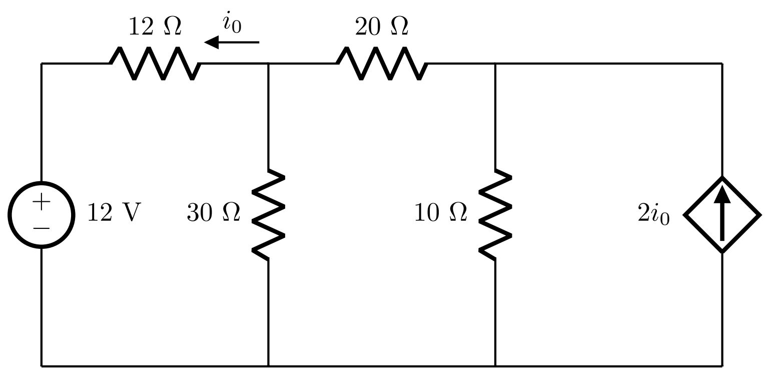
Total Section 1 / Totaal Afdeling 1: [31]

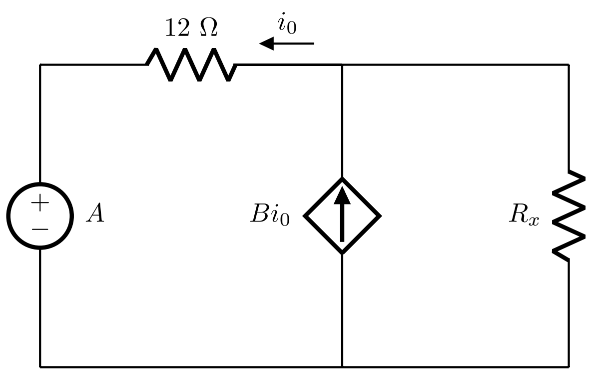
Section 2 / Afdeling 2

Questions 28 to 29 [4]		Vrae 28 tot 29 [4]	
Find the contribution to the current i_x due to:		Bepaal die bydrae tot die stroom i_x as gevolg van:	
28	the 30 V source [2]	**28**	die 30 V bron [2]
29	the 3 A source [2]	**29**	die 3 A bron [2]

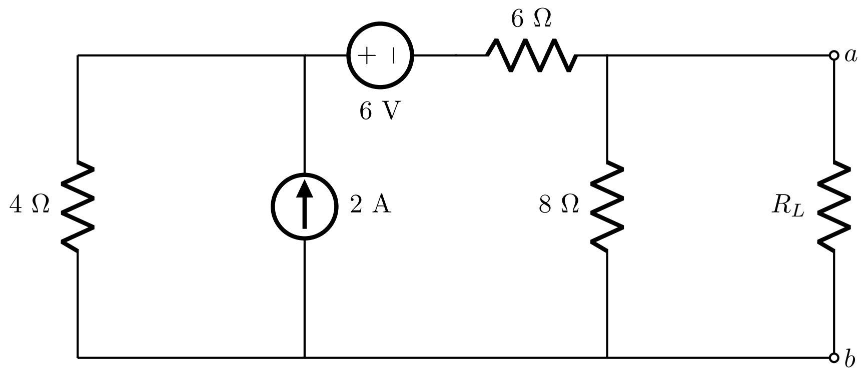


Questions 30 to 32 [5]		Vrae 30 tot 32 [5]	
Simplify the top circuit by means of source transformation so that it looks like the bottom circuit. Determine:		Vereenvoudig die boonste stroombaan met behulp van brontransformasie sodat dit lyk soos die onderste stroombaan. Bepaal:	
30	R_x [2]	**30**	R_x [2]
31	A [1]	**31**	A [1]
32	B [2]	**32**	B [2]

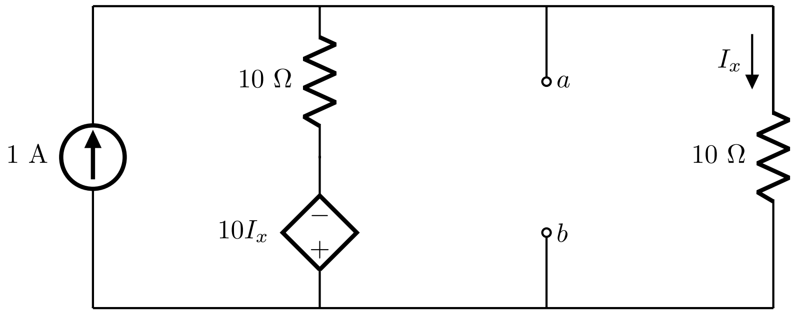




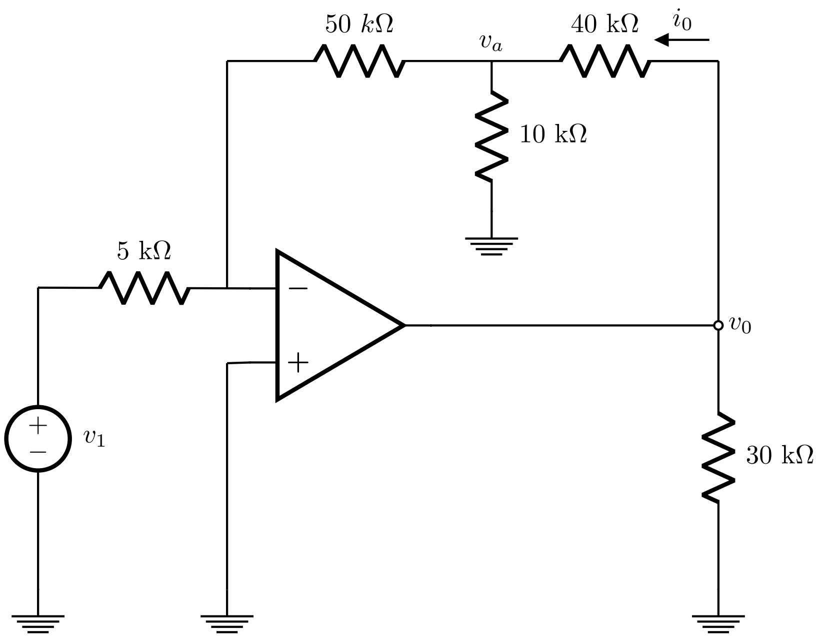
Questions 33 to 34 [4]		**Vrae 33 tot 34 [4]**	
Determine the Norton equivalent current I_N with reference to terminals $a-b$, and the value of the load resistor R_L so that maximum power is transfered to R_L.		Bepaal die Norton ekwivalente stroom I_N met verwysing tot terminale $a-b$, en die waarde van die lasweerstand R_L sodat maksimum drywing oorgedra word aan R_L.	
33	I_N [2]	**33**	I_N [2]
34	R_L [2]	**34**	R_L [2]



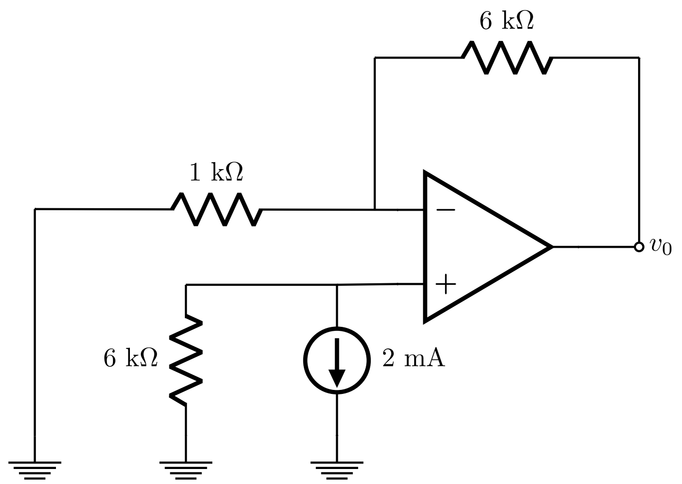
Questions 35 to 36 [4]		Vrae 35 tot 36 [4]	
Find the Thevenin equivalent voltage V_{Th} and the Thevenin equivalent resistance R_{Th} with respect to terminals $a - b$.		Bepaal die Thevenin ekwivalente spanning V_{Th} en die Thevenin ekwivalente weerstand R_{Th} met betrekking tot terminale $a - b$.	
35	V_{Th} [2]	**35**	V_{Th} [2]
36	R_{Th} [2]	**36**	R_{Th} [2]



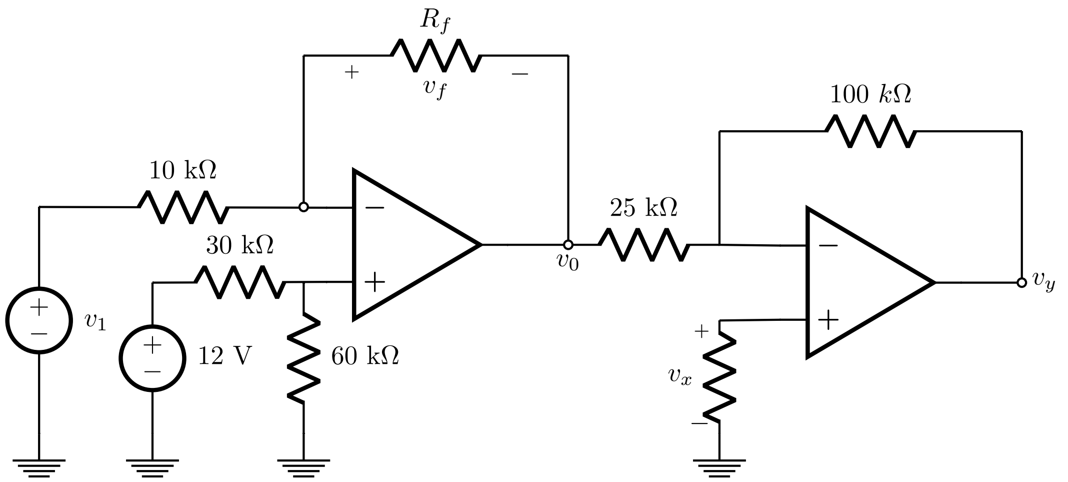
Questions 37 to 39 [6]		**Vrae 37 tot 39 [6]**	
Assume that the op-amp is ideal. It is known that v_a=-0.5 V. Find v_1, v_0 and i_0.		Aanvaar dat die OV ideaal is. Dit is bekend dat v_a=-0.5 V. Bepaal v_1, v_0 en i_0.	
37	v_1 [2]	**37**	v_1 [2]
38	v_0 [2]	**38**	v_0 [2]
39	i_0 [2]	**39**	i_0 [2]



Question 40 [2]		**Vraag 40 [2]**	
Assume that the op-amp is ideal. Find v_0.		Aanvaar dat die OV ideaal is. Bepaal v_0.	
40	v_0 [2]	**40**	v_0 [2]



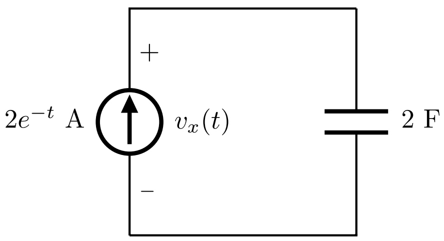
Questions 41 to 43 [6]		**Vrae 41 tot 43 [6]**	
Assume that the op-amps are ideal. It is known that $v_0 = 0$ when $v_1 = v_2$.		Aanvaar dat die OVs ideaal is. Dit is bekend dat $v_0 = 0$ as $v_1 = v_2$.	
41	Find R_f. [2]	**41**	Bepaal R_f. [2]
42	Assume that $v_0 = 20$ V. Find v_f. [2]	**42**	Aanvaar dat $v_0 = 20$ V. Bepaal v_f. [2]
43	Assume that $v_0 = 20$ V. Find v_y. [2]	**43**	Aanvaar dat $v_0 = 20$ V. Bepaal v_y. [2]

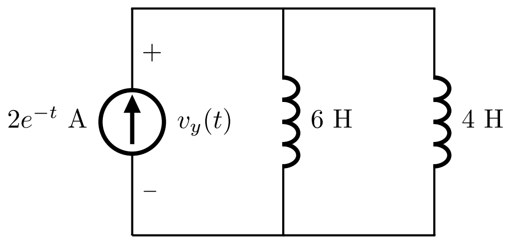


Total Section 2 / Totaal Afdeling 2: [31]

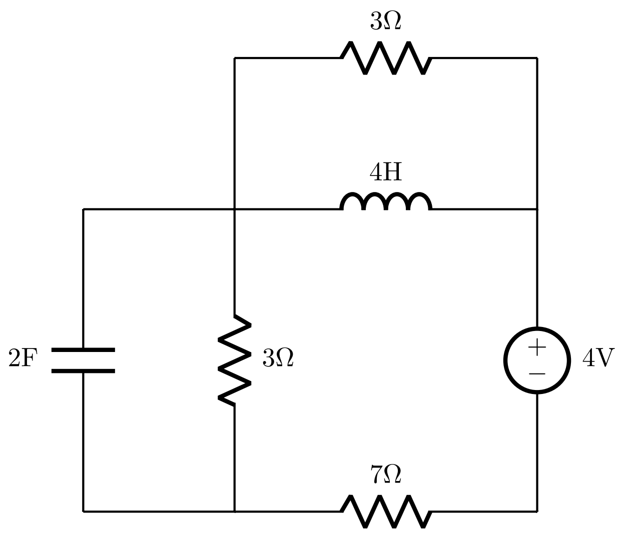
Section 3 / Afdeling 3

Questions 44 to 45 [4]		**Vrae 44 tot 45 [4]**	
Calculate the voltages $v_x(t = 1s)$ and $v_y(t = 1s)$. Assume that $v_x(0) = 0$ V and $v_y(0) = -4.8$ V.		Bereken die spannings $v_x(t = 1s)$ en $v_y(t = 1s)$. Aanvaar $v_x(0) = 0$ V en $v_y(0) = -4.8$ V.	
44	$v_x(t = 1$ s) [2]	**44**	$v_x(t = 1$ s) [2]
45	$v_y(t = 1$ s) [2]	**45**	$v_y(t = 1$ s) [2]

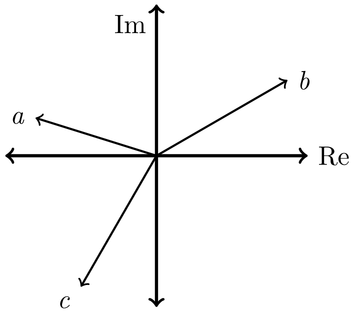




Questions 46 to 47 [3]		**Vrae 46 tot 47 [3]**	
Under direct current conditions, calculate the energy stored in the capacitor (w_C) and the energy stored in the inductor (w_L).		Onder gelykstroomtoestande, bepaal die energie gestoor in die kapasitor (w_C) en die energie gestoor in die induktor (w_L).	
46	w_C [2]	**46**	w_C [2]
47	w_L [1]	**47**	w_L [1]



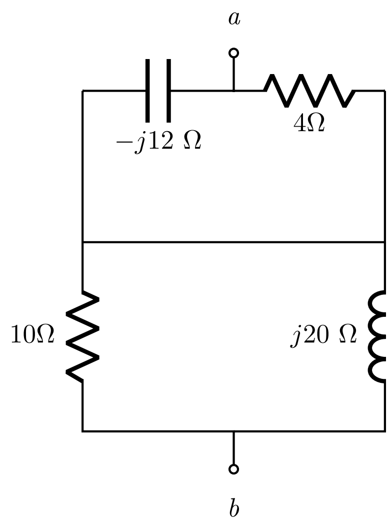
Question 48 [2]		Vraag 48 [2]	
Given the diagram below, which statement is correct? 1. Phasor b lags a and lags c. 2. Phasor c lags b and leads a. 3. Phasor c leads b and leads a. 4. Phasor b leads a and lags c. (Write down the number of the correct statement as your answer.)		Gegewe onderstaande diagram, watter stelling is korrek? 1. Fasor b loop na a en na c. 2. Fasor c loop na b en voor a. 3. Fasor c loop voor b en voor a. 4. Fasor b loop voor a en na c. (Skryf die nommer neer van die korrekte stelling as jou antwoord.)	
48	Statement #:	**48**	Stelling #:



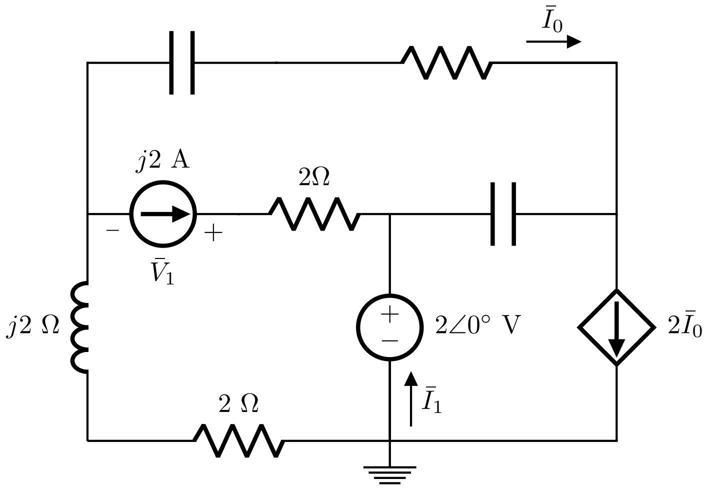
Question 49 [2]		Vraag 49 [2]	
The following sinusoids are given: $v_1(t) = -5\sin(\omega t - 60°)$ V $v_2(t) = \ \ 4\cos(\omega t - 180°)$ V $v_3(t) = \ \ 3\sin(\omega t + 60°)$ V $\bar{V}_1$, $\bar{V}_2$ and $\bar{V}_3$ are the phasor representations of $v_1(t)$, $v_2(t)$ and $v_3(t)$ respectively. Determine the phasor value of $\bar{V}_T = \bar{V}_1 + \bar{V}_2 + \bar{V}_3$. Give the answer in **polar form**.		Die volgende sinusiede word gegee: $v_1(t) = -5\sin(\omega t - 60°)$ V $v_2(t) = \ \ 4\cos(\omega t - 180°)$ V $v_3(t) = \ \ 3\sin(\omega t + 60°)$ V $\bar{V}_1$, $\bar{V}_2$ and $\bar{V}_3$ is die fasorvoorstellings van $v_1(t)$, $v_2(t)$ and $v_3(t)$ respektiewelik. Bepaal die fasor waarde van $\bar{V}_T = \bar{V}_1 + \bar{V}_2 + \bar{V}_3$. Gee die antwoord in **polêre vorm.**	
49	$\bar{V}_T$	**49**	$\bar{V}_T$

Question 50 [2]		**Vraag 50 [2]**	
A series circuit consists of a $40\cos\left(5\times10^3 t\right)$ V voltage source, a 40 Ω resistor, a 2.5 μF capacitor, and 24 mH inductor. The current $i_T(t)$ exits the positive terminal of the voltage source. $\bar{I}_T$ is the phasor representation of $i_T(t)$. Calculate the current $\bar{I}_T$. Give the answer in **rectangular form**.		‘n Serie-kringbaan bestaan uit ‘n $40\cos\left(5\times10^3 t\right)$ V spanningsbron, ‘n 9 kΩ resistor, ‘n 2.5 μF kapasitor, en ’n 24 mH induktor. Die stroom $i_T(t)$ vloei uit die positiewe terminaal van die stroombron. $\bar{I}_T$ is die fasorvoorstelling van $i_T(t)$. Bepaal die stroom $\bar{I}_T$. Gee die antwoord in **reghoekige vorm**.	
50	$\bar{I}_T$	**50**	$\bar{I}_T$

Question 51 [2]		**Vraag 51 [2]**	
Determine Z_{ab}. Give the answer in **rectangular form**.		Bepaal Z_{ab}. Gee die antwoord in **reghoekige vorm**.	
51	Z_{ab}	**51**	Z_{ab}

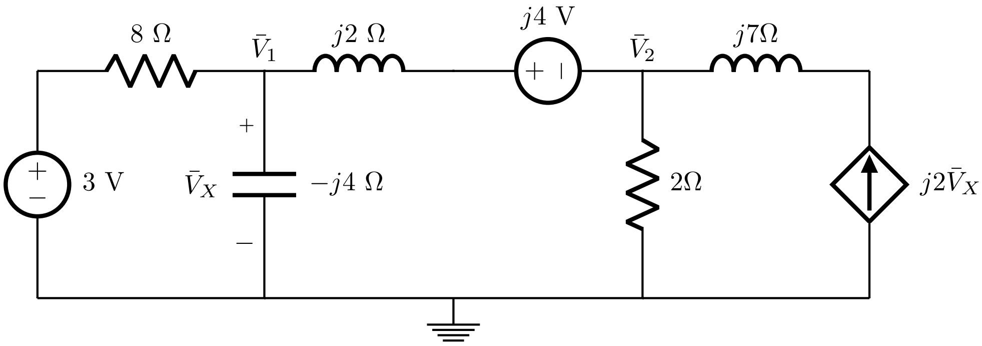


Questions 52 to 53 [4]		**Vrae 52 tot 53 [4]**	
Given: $\bar{I}_0 = 1\angle 90°$ A. Determine $\bar{V}_1$ and $\bar{I}_1$. Give the answers in **rectangular form**.		Gegee: $\bar{I}_0 = 1\angle 90°$ A. Bepaal $\bar{V}_1$ en $\bar{I}_1$. Gee die antwoorde in **reghoekige vorm**.	
52	$\bar{V}_1$	**52**	$\bar{V}_1$
53	$\bar{I}_1$	**53**	$\bar{I}_1$

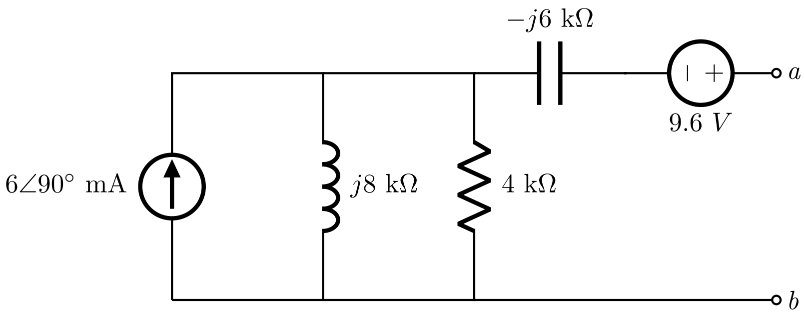


Questions 54 to 57 [4]	Vrae 54 tot 57 [4]
Use nodal analysis to set up the following two equations of the form: $a\bar{V}_1 + b\bar{V}_2 = j19$, $c\bar{V}_1 + d\bar{V}_2 = -j4$. Calculate the unknown coefficients and write in rectangular format.	Gebruik nodale analise om die volgende twee vergelykings op te stel met die vorm: $a\bar{V}_1 + b\bar{V}_2 = j19$, $c\bar{V}_1 + d\bar{V}_2 = -j4$. Bepaal die onbekende koëffisiënte en skryf in reghoekige formaat.

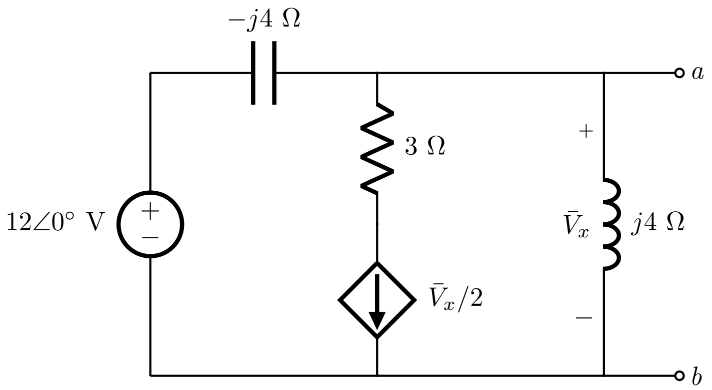
54	a [1]	**55**	b [1]	**56**	c [1]	**57**	d [1]



<table>
<tr><th colspan="2">Questions 58 to 59 [4]</th><th colspan="2">Vrae 58 tot 59 [4]</th></tr>
<tr><td colspan="2">Determine the Thevenin equivalent voltage and impedance for the circuit below with respect to terminals a-b. Give the answers in rectangular form.</td><td colspan="2">Bepaal die Thevenin ekwivalente spanning en impedansie vir onderstaande kringbaan met verwysing tot terminale a-b. Gee die antwoorde in reghoekige vorm.</td></tr>
<tr><td>58</td><td>$\bar{V}_{TH}$</td><td>58</td><td>$\bar{V}_{TH}$</td></tr>
<tr><td>59</td><td>Z_{TH}</td><td>59</td><td>Z_{TH}</td></tr>
</table>



Questions 60 to 61 [4]		Vrae 60 tot 61 [4]	
Determine the Norton equivalent current and impedance for the circuit below with respect to terminals *a*-*b*. Give the answers in **rectangular form**.		Bepaal die Norton ekwivalente stroom en impedansie vir onderstaande kringbaan met verwysing tot terminale *a*-*b*. Gee die antwoorde in **reghoekige vorm**.	
60	$\bar{I}_N$	**60**	$\bar{I}_N$
61	Z_N	**61**	Z_N



Total Section 3 / Totaal Afdeling 3: [31]

Grand Total / Groottotaal : [93]

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	8	E	B	N	1	2	2	E	1		

How to enter complex numbers:										
5	∠	-	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	-	j	8						
j	1	5							Ω	
-	1	.	5	+	j	8	.	2	m	V

Rectangular:	j is at the start of the imaginary number.
	The j is in the CENTRE of the cell.
	The j does NOT cross or touch cell boundaries!
Polar:	The degree sign ° is directly after the angle value.

Please round all answers to a minimum of 3 significant digits.

	Answer	Unit
	SECTION 1	
01	$p =$	
02	$w =$	
03	$p_1 =$	
04	$p_2 =$	
05	$n =$	No unit required
06	$b =$	No unit required
07	$l =$	No unit required
08	$I =$	
09	$v_0 =$	
10	$R =$	
11	$I =$	
12	$R_{ab} =$	
13	$a =$	No unit required
14	$b =$	No unit required
15	$c =$	No unit required
16	$d =$	No unit required
17	$V_{CE} =$	

18	$e =$	No unit required
19	$f =$	No unit required
20	$g =$	No unit required
21	$h =$	No unit required
22	$k =$	No unit required
23	$l =$	No unit required
24	$p =$	No unit required
25	$q =$	No unit required
26	$I_o =$	
27	$v_{out} =$	

SECTION 2

28	30 V source:	
29	3 A source:	
30	$R_x =$	
31	$A =$	No unit required
32	$B =$	No unit required
33	$I_N =$	
34	$R_L =$	
35	$V_{Th} =$	
36	$R_{Th} =$	
37	$v_1 =$	
38	$v_0 =$	
39	$i_0 =$	
40	$v_0 =$	
41	$R_f =$	
42	$v_f =$	
43	$v_y =$	

SECTION 3		
44	$v_x(1\ s) =$	
45	$v_y(1\ s) =$	
46	$w_C =$	
47	$w_L =$	
48	Statement #:	No unit required.
49	$\bar{V}_T =$	
50	$\bar{I}_T =$	
51	$Z_{ab} =$	
52	$\bar{V}_1 =$	
53	$\bar{I}_1 =$	
54	$a =$	No unit required.
55	$b =$	No unit required.
56	$c =$	No unit required.
57	$d =$	No unit required.
58	$\bar{V}_{TH} =$	
59	$Z_{TH} =$	
60	$\bar{I}_N =$	
61	$\bar{Z}_N =$	